Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/18/2006
PI Name:	Ware, Jeffrey Hart Ph.D.		
Project Title:	The Role Of Free Radicals In Space Radiation-Induced Malignant Transformation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6072	**Congressional District:**	2
Comments:	PI is deceased as of 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	09/01/2004	**End Date:**	08/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK46G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposed research investigates the role of oxidative stress in space radiation induced carcinogenesis in vitro and evaluates the efficacy of antioxidants as countermeasures against space radiation induced malignancy. The main hypothesis to be tested in the proposed study is that space radiation induces malignant transformation by increasing oxidative stress, and that prevention of oxidative stress during and after the radiation exposure can prevent down-stream events, thereby suppressing space radiation induced carcinogenesis. The radiation sources to be used in the proposed study are high energy protons and high-energy, highly charged particles known as HZE particles, which are the main components of space radiation of concern for the health of astronauts. The cell system to be used in the study is the HTori-3 human thyroid epithelial cell line that is non-malignant but can be transformed by ionizing radiation to yield cell clones capable of anchorage independent growth in soft agar and tumor formation in athymic nude mice. We plan to over-express four major antioxidant enzymes, i.e., manganese superoxide dismutase (MnSOD), copper-zinc superoxide dismutase (CuZnSOD), catalase (CAT) and glutathione peroxidase (GPX-1), in HTori-3 cells and subject the cells to proton and HZE particle radiation in the presence or absence of small molecular weight antioxidants to determine whether the over-expressed antioxidant enzymes protect the cells against space radiation induced malignant transformation and whether the small molecular weight antioxidants can substitute for antioxidant enzymes or act in synergy with the antioxidant enzymes to protect cells against malignant transformation. We also plan to treat HTori-3 cells with small interfering RNA (siRNA) against the four antioxidant enzymes or with buthionine sulfoximine (BSO) and subject the treated cells to proton and HZE particle radiation to verify the importance of the antioxidant enzymes and glutathione (GSH) in the baseline defense against space radiation induced malignant transformation. This proposed study is expected to yield valuable information to ascertain the role of oxidative stress in space radiation induced carcinogenesis and the feasibility of modulating antioxidant enzyme activity and/or diet supplementation with small molecular weight antioxidants as countermeasures against space radiation induced malignancy. The main hypothesis to be tested in the proposed study is that space radiation induces malignant transformation by increasing oxidative stress, and that prevention of oxidative stress during and after the radiation exposure can prevent down-stream events, thereby suppressing space radiation induced carcinogenesis. This proposed study is expected to yield valuable information to ascertain the role of oxidative stress in space radiation induced carcinogenesis and the feasibility of modulating antioxidant enzyme activity and/or diet supplementation with small molecular weight antioxidants as countermeasures against space radiation induced malignancy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the role of oxidative stress in carcinogenesis--and methods of modulating carcinogenesis rates by modulating oxidative stress--can yield great benefits in the prevention of cancer in those exposed to elevated levels of oxidative stress. Such situations may include people exposed to ionizing radiations such as victims of radiation terrorism (i. e., a dirty bomb) and patients receiving radiation therapy, as well as people exposed chemical oxidants (e. g., smokers.)
Task Progress:	We have generated HTori-3 derivative cell lines using a neomycin-resistant selectable marker and obtained HTori-3 cells stably expressing MnSOD and catalase genes, as well a control clone, which contains the vector without an inserted transcript. These clones have been expanded and many vials frozen to maintain consistency of the cells in the model. We will continue to generate HTori-3 derived cell lines expressing CuZnSOD and GPX during the coming year. We have been using the derived cell lines expressing MnSOD (referred to as HT-Mn) and catalase (referred to as HT-CAT) and the control cell line (referred to as HT-Neo) for several months now. The plating efficiencies and growth rates for the derived cell lines are observed to be variable when compared to the parental HTori-3 cell line. It is presumed that this is due to serum factors, as different lots of serum have produced different results. We are in the process of testing lots of serum to procure a large amount of a favorable lot. Preliminary experiments with these cells show no improvement in survival after irradiation in cells transfected with antioxidant enzymes (HT-Mn and HT-CAT) compared to the parental line (HTori-3), but do show a favorable trend compared to the line transfected with the empty vector (HT-Neo). Expanded experiments are planned to confirm this trend. Malignant transformation of these cells as measured by anchorage independent colony formation has also been erratic. Presumably, this is also due to serum factors, and reproducibility will improve once a favorable lot of serum is identified and secured.
Bibliography Type:	Description: (Last Updated: 12/08/2009)
Abstracts for Journals and Proceedings	Ware JH, Stewart J, Guan J, Zhou Z, Wan XS, St Clair DK, Kennedy AR. "The role of free radicals in space radiation-induced malignant transformation." Bioastronautics Investigators' Workshop, Galveston, Texas, January 2006. Bioastronautics Investigators' Workshop proceedings, January 2006. , Jan-2006
Abstracts for Journals and Proceedings	Ware JH, Stewart J, Guan J, Zhou Z, Wan XS, St Clair DK, Kennedy AR. "The role of free radicals in space radiation-induced malignant transformation." 16th Annual NASA Space Radiation Investigators' Workshop, Port Jefferson NY, June 2005. 16th Annual NASA Space Radiation Investigators' Workshop, June 2005. , Jun-2005